

MOHAMMED NABEEL N H

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PROFILE

B.E. student in Computer Science (Artificial Intelligence) with hands-on experience in Generative AI, Computer Vision, Robotics, and Autonomous Systems. Skilled in PyTorch, Large Language Models (LLMs), Agentic AI, Explainable AI, and Retrieval-Augmented Generation (RAG). Passionate about building AI-powered systems and autonomous technologies that solve real-world problems.

EDUCATION

Maharaja Institute of Technology Mysore

Bachelor of Engineering in Computer Science (Artificial Intelligence)

Mysore, India

2023 – 2027

Pre-University Education Board, Government of Karnataka

GopalaSwamy Shishuvihara Institutions (Science Stream)

Mysore, India

2021 – 2023

TECHNICAL SKILLS

Programming Languages: Python | C | JavaScript

Web Development: HTML | CSS | React.js | Next.js

ML/DL Frameworks: PyTorch | TensorFlow | Keras | Scikit-learn | LangChain | LangGraph

Backend & Deployment: Flask | FastAPI | Django | Django REST Framework | REST APIs | Docker | Git

Databases: MongoDB | SQLite | Firebase | Supabase | postgresql

Core AI Areas: Machine Learning | Deep Learning | NLP | Generative AI | Computer Vision | CNN Architectures | Agentic AI

Applied AI Skills: Model Training | Fine-Tuning | Retrieval-Augmented Generation | Prompt Engineering | Explainable AI

PROJECTS

Krishi Sakhi V2.0 – AI-Powered Digital Farming Assistant | Python, Django, DRF, React.js,

TensorFlow/Keras, Scikit-learn, Gemini API

2025

- Built a full-stack AI platform for crop recommendation, plant disease detection, and farmer assistance using Django, React.js, and TensorFlow.
- Developed a CNN-based disease detection model achieving 81% test accuracy and integrated Random Forest-based crop recommendation using soil and weather data with geolocation-based recommendations.
- Integrated a Gemini-powered agriculture chatbot and secure REST APIs with JWT authentication for intelligent farmer support.

Fetal Head Segmentation with Explainable AI (XAI) | Python, PyTorch, U-Net, OpenCV, Grad-CAM, Docker 2026

- Developed and Integrated Explainable AI (XAI) techniques such as Grad-CAM to visualize model predictions and provide interpretable insights into the segmentation results.
- Evaluated model performance using Dice Score, IoU, Precision, and Recall, with an interactive interface for uploading ultrasound images and visualizing segmentation outputs.

EduTrack – College ERP & Academic Management System | Flutter, Dart, Firebase

2025

- Developed a multi-role college ERP application for students, teachers, and administrators with secure role-based authentication.

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- Implemented attendance tracking, marks management, and automated academic performance calculation systems.
 - Built dashboard interfaces and scalable backend APIs for efficient academic data management and analytics.

Future Grid — Artificial Society Simulation System): | Python, FastAPI, Next.js, React, TypeScript, Google Gemini API, Multi-Agent Systems (MAS), LLM Orchestration, Zustand, Tailwind CSS, REST APIs. 2026

- Multi-Agent Simulation Platform: Built an AI-driven society simulation engine modeling 500+ autonomous agents across 9 socio-economic classes to predict human decision-making, product adoption, and market reactions.
- Episodic Memory & Orchestration Pipeline: Architected an asynchronous backend engine managing agent lifecycles, event broadcasting, and decay-weighted episodic memory using LLM integration for realistic behavior.
- Real-Time Interactive Analytics Dashboard: Developed a responsive data visualization dashboard featuring live decision funnels, geospatial agent cluster tracking, and real-time sentiment metrics.

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 – Generative AI Professional | Oracle University

Oracle Cloud Infrastructure – AI Foundations Associate | Oracle University

IBM Artificial Intelligence Fundamentals | IBM SkillsBuild

IBM Data Science Fundamentals with Python & SQL | Coursera

Ethical Hacking | IEEE

ACHIEVEMENTS & EXTRACURRICULAR ACTIVITIES

Student Ambassador –LetsUpgrade | LetsUpgrade 2025

- Selected as a Student Ambassador for LetsUpgrade, contributing to student engagement, promoting learning initiatives, and connecting peers with technology-focused educational opportunities..

National – Level Conference Presentation | MIT First Grade College 2026

- Authored and presented a research paper titled “Refurbished_shopping-agen” at the National-Level Conference on Frontiers in AI-Based Applications.

PUBLICATION – Indian Patent Office | Indian Patent Office

- AUTONOMOUS MICRO-DRONE SYSTEM FOR REAL-TIME POWER LINE INSPECTION, MICRO-CRACK DETECTION AND ON-SPOT CONDUCTIVE REPAIR, Indian Patent Office – Application No:202541117134 2025